Zürich, Switzerland & Oslo, Norway
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Xiangtian Shi

Education

2026 – Present **ETH Zürich, Switzerland**
M.Sc. in Robotics, Systems and Control
Current coursework: Model Predictive Control, Robot Dynamics, Planning and Decision Making for Autonomous Robots, Vision Algorithms for Mobile Robotics.

2023 – 2026 **KTH Royal Institute of Technology, Stockholm, Sweden**
B.Sc. in Vehicle Engineering (GPA: 4.9/5.0)
Selected coursework: Automatic Control, Optimization, Numerical Methods, Applied Computer Science, Signals and Mechanical Systems, Electrical Engineering.

Technical Skills

Programming Python, MATLAB, Arduino (C++)

Tools PyTorch, Git, Solid Edge, ANSYS, RocketPy

Engineering Control systems, modeling and simulation, embedded systems, mechanical design, hardware integration, experimental setup design, data acquisition

Technical Experience and Projects

09.2026 – Present **Real-World Robotics: Articulated Robotic Hand**
Project Course, ETH Zürich, Switzerland
- Working in a Linux and ROS 2 environment on reinforcement-learning policy training for an articulated robotic hand.
- Working on long-horizon task learning and evaluation on the physical robot.

06.2026 – Present **Biomimetic Flapping-Wing Robot Sim-to-Real Research**
Visiting Student Researcher, Dr. Tao Du's Group, IIIS, Tsinghua University, China
- Developed an ESP32-based wireless SBUS control interface for the group's flapping-wing research; integrated and troubleshot experimental hardware.
- Redesigned mechanical components for motion-capture and force-measurement data acquisition to improve measurement reliability and support sim-to-real validation.
- Contributed to simulation workflows linking model development to physical experiments.

01.2026 – 06.2026 **PINNs for Optimal Trajectory Planning of Unicycle Robots**
Bachelor's Thesis, KTH, Sweden
- Developed PyTorch frameworks for physics-informed trajectory optimization under kinematic and control constraints.
- Implemented physics-based constraint enforcement and obstacle-avoidance strategies for unicycle trajectory planning.
- Evaluated loss balancing and constraint enforcement across the optimization workflow.

01.2025 – 09.2025 **Rocket Avionics Engineer**
AESIR Student Rocketry Association, KTH, Sweden
- Developed a ballistic-coefficient apogee predictor maintaining $\leq$0.6% error after settling ($\sim$1.5 s post-burnout), vs. up to 4.2% for the drag-table baseline in RocketPy.
- Co-developed a PID airbrake controller for the Freya rocket with 100 Hz apogee prediction, achieving 0.2% apogee error in nominal RocketPy simulations.

01.2024 – 06.2024 **Solar-Powered Model Vehicle**
Project Course, KTH, Sweden
- Designed and built a vehicle integrating solar power, mechanical components, and electrical subsystems.
- Implemented Arduino-based control and evaluated the integrated system through performance testing.

Work Experience

10.2025 – 06.2026 **Teaching Assistant in Electrical Engineering**
Unit of Mechatronics, KTH, Sweden
- Supported laboratory instruction in circuit theory, signal processing, and mechatronics.
- Evaluated assignments and assisted with course coordination.

05.2023 – 01.2025 **Mathematics Tutor**
MentorNorge AS, Norway
- Delivered individualized mathematics tutoring adapted to each student's learning needs.

10.2022 – 06.2023 **Kindergarten Assistant**
Høvik kindergarten, Norway
- Supported children aged 2–4 in daily routines and structured creative and educational activities.

Languages

Norwegian (Native); Chinese (Native); English (Fluent; IELTS Academic 7.5); Swedish (Fluent; CEFR B2)

Extracurricular Activities

09.2023 – 06.2024 **International Group**
KTH Student Council, Sweden
- Organized social and cultural events supporting integration among international students.

10.2020 – 08.2022 **Competitive Esports**
Professional PUBG Mobile Player
- Won the PUBG Mobile Pro League Western Europe Spring 2021 in a 16-team professional field and competed internationally.

Hobbies and Interests

Violin (10+ years, regional competition winner); Table tennis (club member, regional competitions); Running (Stockholm Half Marathon 2025); Hiking; Skiing